SQL EXERCISE 2

Question 1

```
SELECT COUNT(id) AS total_employees
FROM employees
```

| | total employees |
|---|---|
| 1 | 10 |

Question 2

```
SELECT SUM(salary) AS total_salary
FROM employees
WHERE department = 'IT'
```

| | total salary |
|---|---|
| 1 | 220000 |

Question 3

~~SE~~

```
SELECT AVG(salary)
FROM employees
WHERE department = 'HR'
```

| | AVG(salary) |
|---|---|
| 1 | 49500 |

Question 4

```
SELECT MAX(~~sum~~salary), MIN(salary)
FROM employees;
```

| | max(salary) | min(salary). |
|---|---|---|
| 1 | 62000 | 48000 |

Question 5

```
SELECT SUM(salary)
FROM employees
GROUP BY department;
```

| | Sum Salary |
|---|---|
| 1 | 220 000 |
| 2 | 99000 |
| 3 | 119666 |
| 4 | 105000 |

Question 6

```
SELECT city, COUNT(salary) AS employee_count
FROM employees
GROUP BY city;
```

| | city | employee_count |
|---|---|---|
| 1 | New York | 2 |
| 2 | Chicago | 3 |
| 3 | Los Angeles | 2 |
| 4 | San Francisco | 2 |
| 5 | Houston | 1 |

Question 7

```
SELECT AVG(salary), department
FROM employees
GROUP BY department
ORDER BY AVG(salary) DESC;
```

| | AVG(salary) | department |
|---|---|---|
| 1 | 59833 | Finance |
| 2 | 55000 | IT |
| 3 | 52500 | Marketing |
| 4 | 49500 | HR |

Question 8

```
SELECT department
       SUM(salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100000
```

| | department | total_salary |
|---|---|---|
| 1 | IT | 220000 |
| 2 | Finance | 119666 |
| 3 | Marketing | 105000 |

Question 9

```
SELECT COUNT(id) AS count_id,city
FROM employees
GROUP BY city
HAVING count_id > 1
ORDER BY count_id DESC;
```

| | count_id | city |
|---|---|---|
| 1 | 3 | Chicago |
| 2 | 2 | New York |
| 3 | 2 | Los Angeles |
| 4 | 2 | San Francisco |

Question 10

```
SELECT department, MAX(salary) AS average_salary
FROM employees
GROUP BY department;
```

| | department | average_salary |
|---|---|---|
| 1 | IT | 62000 |
| 2 | HR | 51000 |
| 3 | finance | 60 666 |
| 4 | Marketing | 53000 |